Then, the mean of his scores (that is the arithmetic average of the scores) is obtained by

$$\bar{X} = \frac{\sum x}{n} = \frac{25 + 32 + 36 + 38 + 45 + 41 + 35 + 36}{8} = \frac{288}{8} = 36 .$$

In general, if we have n number of observations $x_1, x_2, x_3, \ldots, x_n$ then their arithmetic mean denoted by $\bar{X}$ (read as X bar) is given by

$$\bar{X} = \frac{\text{sum of all the observations}}{\text{number of observations}} \quad \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

We express this as a formula: $\bar{X} = \frac{\sum x}{n}$

Assumed Mean method: Sometimes we can make calculations easy by working from an entry that we guess to be the right answer. This guessed number is called the assumed mean.

In the example above on cricket scores, let us assume that 38 is the assumed mean. We now list the differences between the assumed mean and each score entered:

25–38 = –13, 32–38 = –6, 36–38 = –2, 38–38 = 0,

45–38 = 7, 41–38 = 3, 35–38 = –3, 36–38 = –2

The average of these differences is $\frac{-13-6-2+0+7+3-3-2}{8} = \frac{-16}{8} = -2$

We add this 'mean difference' to the assumed mean to get the correct mean.

Thus the correct mean = Assumed Mean +Mean difference = 38 – 2 = 36 .

This method will be very helpful when large numbers are involved.

Note

It does not matter which number is chosen as the assumed mean; we need a number that would make our calculations simpler. Perhaps a choice of number that is closer to most of the entries would help; it need not even be in the list given.

8.4.2 Arithmetic Mean-Ungrouped Frequency Distribution

Consider the following list of heights (in cm) of 12 students who are going to take part in an event in the school sports.

140, 142, 150, 150, 140, 148, 140, 147, 145, 140, 147, 145.

How will you find the Mean height?

There are several options.

(i) You can add all the items and divide by the number of items.

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